Closed-Book Knowledge Base Construction with Calibrated Abstention and Numeric Self-Consistency

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Abstract

Large language models (LMs) store a remarkable amount of factual knowledge, yet turning it into a reliable, *set-valued* knowledge base, where a subject may have zero, one, or many objects, remains difficult. The LM-KBC 2026 shared task (EMNLP 2026) poses this in a strict *closed-book* setting: an open-weight model of at most 32B parameters, with no retrieval, no external knowledge, and no fine-tuning. We start from the evaluation metric and observe that it rewards *abstention*: because macro-F_1 scores an empty prediction against an empty gold set as perfect, predicting nothing everywhere already reaches 0.175 relation-averaged macro-F_1, close to the official baseline (0.273). We therefore treat *when to answer* as a first-class decision and build a per-relation pipeline that grounds each prompt in the relation's exact scope, decides abstention from the generated set rather than by instruction, and aggregates numeric relations by median. With **Gemma-4-31B** (a 30.7B open-weight model, the strongest in the budget) a single sample per instance reaches 0.459 relation-averaged macro-F_1 (0.490 micro), beating the official baseline (0.273/0.284) on all six relations; decode-time self-consistency voting on a smaller 4-bit Qwen2.5-14B reaches only 0.416, showing that model scale outweighs voting at this budget. All generation is cached, so the decode stage is reproducible without a GPU. We release our code.[1]

1 Introduction

Pre-trained language models (LMs) store a remarkable amount of factual knowledge in their parameters, acquired during pre-training on large text corpora (Petroni et al., 2019; Roberts et al., 2020), and a body of work probes and extracts it (Jiang et al., 2020; Zhong et al., 2021; AlKhamissi et al., 2022). This raises the prospect of building a structured knowledge base (KB) directly from an LM rather than from curated sources (Dong et al., 2014; Vrandečić and Krötzsch, 2014): no schema engineering, no hand annotation, and an open query set. Doing so reliably, however, is harder than answering a single factual question. A KB entry is a *complete set* of facts, so the model must decide not only what is true but how many objects, possibly none, a subject and relation admit.

The LM-KBC 2026 shared task (Alivanistos et al., 2022; Kalo et al., 2023, 2025) formalises exactly this: given a subject s and relation r, predict the complete object set $\{o_1, \ldots, o_k\}$, where k may be zero, one, or many (Section 3). Predictions are scored by macro precision, recall, and F_1, with string normalisation and alias matching for entities and a 5% tolerance for numeric relations. The central difficulty is that this is not single-answer probing: unlike benchmarks such as LAMA (Petroni et al., 2019), which assume exactly one gold object and reward ranking it highly, this setting demands the *entire* set, and the empty set is a legitimate, frequent answer (a living person has no recorded city of death; an island country has no land border). Three difficulties follow. LM outputs are stochastic and prompt-sensitive, so a single query yields an unstable set; relations span very different cardinalities, from a single number to hundreds of award recipients, which complicates aggregation; and the model must recognise when it does not know an answer and *abstain* rather than fabricate one. The 2026 edition compounds the challenge by being strictly *closed-book*: a system may use any open-weight model up to 32B parameters, but no web search, no retrieval, no external corpus, and no fine-tuning. This forecloses the retrieval and tuning that strong entries in earlier editions relied on (Zhang et al., 2023; Asai et al., 2024), leaving parametric recall and prompt design as the

[1] https://github.com/ANON/lm-kbc-2026

only levers.

In this paper, we start from the evaluation metric itself rather than from the model. Because macro-F_1 scores an empty prediction against an empty gold set as perfect, simply predicting nothing everywhere already attains 0.175 relation-averaged macro-F_1 on validation, competitive with the official baseline (0.273) and higher on the most null-heavy relations (Section 3). The lesson is that points come not from answering everything, but from recalling the few numeric values and, above all, from knowing when *not* to answer. We therefore make abstention a first-class decision: for each null-heavy relation we take several samples and keep an object only when the samples agree, collapsing to the empty set when the model is internally inconsistent, while scope-grounded prompts and median-of-samples aggregation handle the remaining relations (Figure 1). We apply this single pipeline to all six relations of the task with one open-weight model, Gemma-4-31B (30.7B, the strongest within the 32B budget), and reuse a cached set of generations so that the entire decode and abstention stage is a deterministic, GPU-free function of that cache.

This paper makes three contributions. First, a metric analysis (Section 3) shows that under the macro metric abstention is a high-value action: the predict-nothing baseline already rivals the official baseline and beats it on the most null-heavy relations, which reframes the task around *when to answer*. Second, we build a per-relation pipeline (Figure 1, Section 4) that grounds prompts in exact scopes and induces *calibrated abstention* via self-consistency voting, with no trained confidence model; it lifts a weak 0.5B model from 0.173 to 0.372 on the null-heavy relations, clearing the predict-nothing floor rather than merely matching it, and still helps a strong 14B. Third, the resulting open-weight system reaches 0.459 relation-averaged macro-F_1 (0.490 micro) with Gemma-4-31B, beating the official baseline (0.273/0.284) on all six relations and clearing the empty baseline (0.175) by a wide margin; a local 4-bit Qwen2.5-14B reaches 0.416/0.447 under the same pipeline, and comparing it against Gemma-4-31B shows that at this budget model scale outweighs decode-time self-consistency (Section 7).

2 Background and Related Work

LMs as knowledge bases. Petroni et al. (2019) cast factual recall as cloze probing; follow-ups sharpened the prompts (Jiang et al., 2020; Shin et al., 2020; Zhong et al., 2021), quantified how much knowledge a model's parameters hold (Roberts et al., 2020), and questioned how reliably it can be read out (Cao et al., 2021; AlKhamissi et al., 2022). These probes assume a single gold object and reward ranking it highly. We instead target the set-valued, alias-graded variant in which the empty set is a valid answer, a setting closer to materialising an actual KB (Pan et al., 2023).

The LM-KBC shared task. Across its editions (Alivanistos et al., 2022; Kalo et al., 2023, 2024, 2025) the task has moved from masked-LM probing and prompt ensembling toward instruction-tuned generative systems, with some earlier tracks permitting retrieval or fine-tuning (Zhang et al., 2023). The 2026 edition's closed-book, no-tuning, $\leq$32B rules remove those options, which is precisely what makes our parametric-recall-with-abstention design necessary.

Retrieval vs. closed book. The usual route to knowledge-intensive accuracy is retrieval augmentation (Lewis et al., 2020; Guu et al., 2020), which helps most exactly when parametric memory is weak (Mallen et al., 2023). Self-RAG (Asai et al., 2024) even lets the model decide *when* to retrieve; we borrow the spirit of a model-driven, adaptive decision but, since retrieval is forbidden, redirect it to a different question: *when to answer at all*, rather than what to fetch.

Self-consistency, calibration, and abstention. These three threads converge in our method. Self-consistency (Wang et al., 2023; Chen et al., 2023) samples a model repeatedly and aggregates the generations to pick a single answer. Abstaining when uncertain is the classic reject option (Geifman and El-Yaniv, 2017), studied in NLP as calibration (Guo et al., 2017), unanswerable QA (Rajpurkar et al., 2018), and selective QA (Cole et al., 2023), and LMs retain some awareness of what they know (Kadavath et al., 2022). We combine the two: cross-sample *disagreement* (Kuhn et al., 2023; Manakul et al., 2023) is our zero-parameter confidence signal, and, unlike majority-vote self-consistency, which always emits an answer, we let it trigger *abstention*, because here the empty set is itself a valid, high-value prediction.

Relation	#val	empty%
COUNTRYLANDBORDERSCOUNTRY	68	25
PERSONHASCITYOFDEATH	100	45
COMPANYTRADESATSTOCKEXCHANGE	100	35
AWARDWONBY	10	0
HASCAPACITY	100	0
HASAREA	100	0

Table 1: Validation statistics. AWARDWONBY averages $\sim$148 objects per subject (max 641); three string relations are often legitimately empty.

3 Task and Metric Analysis

Relations. The dataset comprises six Wikidata relations (Table 1) that fall into three groups. *Numeric* relations (HASCAPACITY, HASAREA) have a single scalar object; *large-set* AWARDWONBY has up to hundreds of recipients per subject; and three *null-heavy* string relations (COUNTRYLANDBORDERSCOUNTRY, PERSONHASCITYOFDEATH, COMPANYTRADESATSTOCKEXCHANGE) leave a large fraction of subjects with no valid object at all. Each relation carries a precise scope (land-only borders, city-granular death location, the highest published capacity), which we copy verbatim into the prompt so the model targets the same notion the gold set was constructed under.

Evaluation. Predictions are scored against the gold object set by macro precision and recall. A predicted string is a true positive if its normalised form (lower-cased, diacritics and punctuation removed) matches any alias of a gold object, and a numeric prediction if it lies within 5% of the gold value. Two boundary conventions drive everything that follows: precision is defined as 1 when the prediction is empty, and recall as 1 when the gold set is empty. We report *relation-averaged* macro-F_1 (the mean of the six per-relation scores) and *micro-averaged* F_1 (over all instances).

The metric makes abstention valuable. Those two conventions make an empty prediction against an empty gold set a perfect F_1, with a large consequence: predicting the empty set for *every* validation instance already scores the "Empty" bars in Figure 3, averaging 0.175 relation-averaged macro-F_1 (0.203 micro). This nearly matches the official baseline's 0.273 relation-averaged macro-F_1 and *beats* it on the two most null-heavy relations (PERSONHASCITYOFDEATH 0.45 vs. 0.16; COMPANYTRADESATSTOCKEXCHANGE 0.35 vs. 0.32). The task therefore rewards two narrow abilities rather than answering everything: recalling the numeric values, and abstaining precisely when no object exists. Our pipeline pulls exactly these two levers.

4 System

Architecture overview. The pipeline (Figure 1) maps each subject–relation pair (s, r) to one chat prompt, decodes a cached set of samples (N for voting, n for the numeric median), and aggregates them per relation kind into an object set. It has no trainable parameters and targets the two analysis levers: abstention and numeric recall.

Scope-grounded prompts. Each relation's system message states the task, the *exact* ground-truth scope (copied verbatim from the official definitions), and the required answer form (Table 2); the model emits one JSON object `{"answers":[...]}` (Figure 2), which makes the empty case unambiguous. Few-shot exemplars come from the training split; for null-heavy relations we interleave empty and non-empty exemplars so abstention is demonstrated, not merely instructed. Even a 0.5B model emits valid JSON on 100% of instances, so format is never a bottleneck.

Calibrated abstention via self-consistency. Scope grounding tells the model *what* counts as an answer, but not *when* to stay silent. A bare instruction to "answer empty when unsure" is unreliable: left unconstrained, a small model names a death-city for living people and invents stock exchanges, dropping precision *below* the empty baseline (Section 7). Instead, for the three null-heavy relations we sample N=3 completions at temperature τ=0.7 and keep an object only if it appears in at least a fraction θ of samples; single-object relations abstain entirely if even the top answer is below θ (Algorithm 1). When the model guesses, samples disagree and the prediction collapses to $\emptyset$, exactly where the metric rewards silence. We disable voting for AWARDWONBY (recall-bound) and the numeric relations.

Numeric aggregation. The numeric relations need a different aggregator: HASCAPACITY and HASAREA take a single number graded within 5%. We sample n=5 completions and take the *median* of the parsed numbers (Wang et al., 2023), which resists outlier hallucinations.

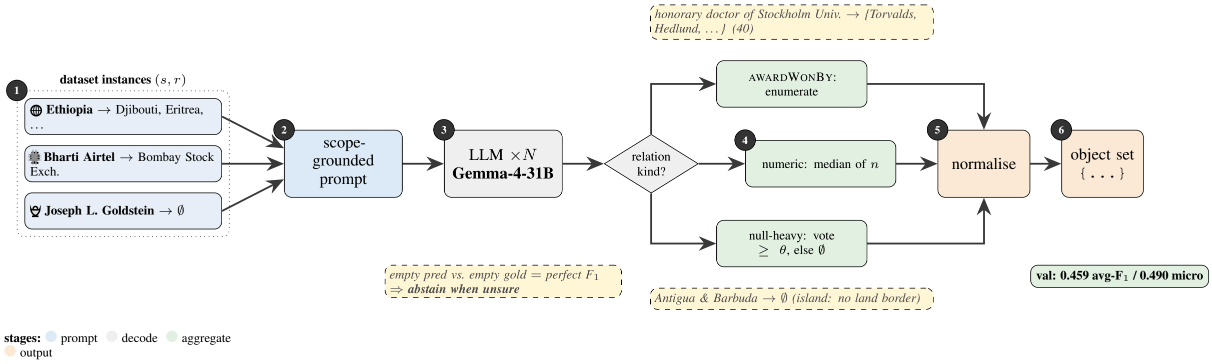

Figure 1: System pipeline (no trainable parameters). Real dataset instances (left, stage 1), shown with entity-type icons and their gold objects, are each rendered into a scope-grounded chat prompt (stage 2); the model is sampled N times (stage 3, cached), and the samples are aggregated by relation kind (stage 4): *median* for numeric relations, consistency *voting with abstention* (keep an object only if it appears in $\geq \theta$ of samples, else $\emptyset$) for the null-heavy string relations, and *enumeration* for AWARDWONBY; a final normalisation matches the evaluator exactly. Dashed boxes are real worked examples: the island country ANTIGUA & BARBUDA correctly collapses to $\emptyset$, as does the still-living scientist JOSEPH L. GOLDSTEIN (no city of death). The design follows from the metric insight (lower-left badge). Reported score is Gemma-4-31B on validation.

Relation	Scope stated in the prompt	Answer form
COUNTRYLANDBORDERSCOUNTRY	countries sharing a *land* border; maritime borders excluded; an island has none	countries, or []
PERSONHASCITYOFDEATH	the *city* where the subject died; empty if alive or the city is unknown	one city, or []
COMPANYTRADESATSTOCKEXCHANGE	stock exchange(s) on which the company is publicly listed; empty if unlisted	exchanges, or []
AWARDWONBY	every recipient of the award (often tens to hundreds)	names (long)
HASCAPACITY	maximum spectator capacity in people; highest published value	one integer
HASAREA	total surface area in km^2 (land + inland water)	one number

Table 2: Per-relation prompt specification. Each system message states the relation's exact scope (copied from the official definitions) and the required answer form; the model replies with a single JSON object `{"answers":[...]}`.

```
SYS: precise KB engine (closed book).
TASK: city where the subject died.
SCOPE: city granularity; [] if
alive/unknown.
Reply only {"answers":[...]}.
U: Egbert Mulder
A: {"answers":["Groningen"]}
U: Dave Keon
A: {"answers":[]}
U: <query>  A:
```

Figure 2: Abbreviated prompt for a null-heavy relation. Mixed exemplars teach abstention; output is forced JSON.

Robust parsing. Whatever the relation kind, every sample passes through a common final stage: a relation-aware parser recovers the answer list from noisy output (code fences, prose), strips explicit empty markers, extracts one number for numeric relations (handling commas, units, scale words), and de-duplicates. Normalisation (lower-casing, diacritic/punctuation removal) is byte-identical to

Algorithm 1 CALIBRATEDABSTENTION

Require: samples $S{=}\{s_1, \ldots, s_N\}$, threshold θ, *multi*
1: $c[o] \leftarrow \#\{s_i : o \in \text{PARSE}(s_i)\}$ for each object o
2: **if** *multi* **then return** $\{o : c[o]/N \geq \theta\}$
3: **else**
4: $\quad o^* \leftarrow \arg\max_o c[o]$; **return** $c[o^*]/N \geq \theta \, ? \, \{o^*\} : \emptyset$
5: **end if**

the official evaluator; pushing gold answers back through the parser scores 1.000, confirming it is lossless.

5 Experimental Setup

Models. Our primary system uses **Gemma-4-31B-it** (Gemma Team, 2024), a 30.7B-parameter open-weight model within the 32B cap. It exceeds the memory of our single 16 GB GPU, so we serve it through a hosted OpenAI-compatible API. The weights are open and we apply no fine-tuning, so the closed-book rules are respected: the API is used purely as compute. For the self-consistency ablations (Section 7) we additionally run **Qwen2.5-14B-Instruct** (Qwen Team, 2024) (14.7B) locally in 4-bit. Several nominally "32B" models in fact exceed the cap (Qwen2.5-32B at 32.5B, Qwen3-32B at 32.8B) and are ineligible.

Decoding. The pipeline selects an aggregator per relation kind (Section 4): the median over n=5 samples for numeric relations, consistency voting over N=3 samples at threshold θ=0.5 for null-heavy relations (Algorithm 1), and a single greedy enumeration for AWARDWONBY. We evaluate these self-consistency configurations (temperature τ=0.7) on Qwen2.5-14B, which we can sample freely. API rate limits preclude multi-sampling Gemma-4-31B at scale, so we decode it with a *single* sample per query, yet it still outperforms the 14B (Section 6), indicating that model scale dominates decode-time self-consistency at this budget. Prompts are 8-shot for every relation except AWARDWONBY (2-shot), with exemplars from the training split (Brown et al., 2020).

Hardware and implementation. Qwen2.5-14B runs on a single NVIDIA T4 (16 GB) via HuggingFace Transformers in 4-bit (`bitsandbytes nf4`, `float16` compute) (Dettmers et al., 2023); quantisation only fits the model to the hardware and does not change the counted parameter budget. Gemma-4-31B is queried over its API from the same machine. Each model processes the full 478-instance validation set.

Baselines. We report two references on the same validation split: the *empty* baseline, which predicts $\emptyset$ for every instance (Section 3), and the *official* organiser baseline, a few-shot prompting system whose per-relation scores we take from the task repository.

Reproducibility. Generation caches every raw sample; all post-processing (parsing, consistency voting, median aggregation, and the threshold sweep) is a deterministic, GPU-free function of that cache. Every ablation in Section 7 therefore comes from a single generation pass and reproduces without a GPU. We release all code and prompts.

Relation	Empty	Base	14B	30B
countryLandBorders	0.250	0.679	0.900	**0.953**
companyTrades	0.350	0.320	0.616	**0.722**
personCityOfDeath	0.450	0.160	**0.480**	0.460
hasArea	0.000	0.310	0.330	0.330
hasCapacity	0.000	0.100	0.090	**0.170**
awardWonBy	0.000	0.069	0.081	**0.120**
Avg. of relations	0.175	0.273	0.416	**0.459**
Micro (all inst.)	0.203	0.284	0.447	**0.490**

Table 3: Validation macro-F_1 by model. "Empty" is the measured predict-nothing baseline, "Base" the official baseline; **30B** is our primary Gemma-4-31B system, 14B is Qwen2.5-14B. Best per row in bold. Gemma-4-31B beats the official baseline on *all six* relations.

6 Results

Our primary system, **Gemma-4-31B**, scores **0.459** relation-averaged macro-F_1 (**0.490** micro) on validation, versus 0.175/0.203 (empty), 0.273/0.284 (official baseline), and 0.416/0.447 for the smaller Qwen2.5-14B. Figure 3 and Table 3 give the per-relation picture. Gemma-4-31B *beats the official baseline on all six relations* and the 14B on four of six; the largest gains are on COMPANYTRADESATSTOCKEXCHANGE ($0.62 \rightarrow 0.72$) and HASCAPACITY ($0.09 \rightarrow 0.17$, nearly doubled), while COUNTRYLANDBORDERSCOUNTRY reaches 0.953. It ties the 14B on HASAREA and trails only on PERSONHASCITYOFDEATH (0.46 vs. 0.48), the sole relation where the single-sample 30B forgoes the abstention voting that the 14B uses. Strikingly, the 30B attains this lead *without* self-consistency or abstention, showing that model scale outweighs these decode-time mechanisms at this budget.

7 Analysis

All ablations are decoded from the same cached samples.

Abstention rescues weak models, refines strong ones. Table 4 isolates abstention on the three null-heavy relations. On a poorly-calibrated 0.5B model the effect is dramatic ($0.173 \rightarrow 0.372$): without voting it over-answers and falls *below* the empty baseline (0.350). The 14B is already above that baseline without voting (0.652), so voting adds less ($0.652 \rightarrow 0.666$) but still helps. Abstention buys precision: on PERSONHASCITYOFDEATH it moves P=$0.77 \rightarrow 0.81 \rightarrow 0.89$ as θ rises, trading a little recall: it suppresses wrong guesses, not correct answers.

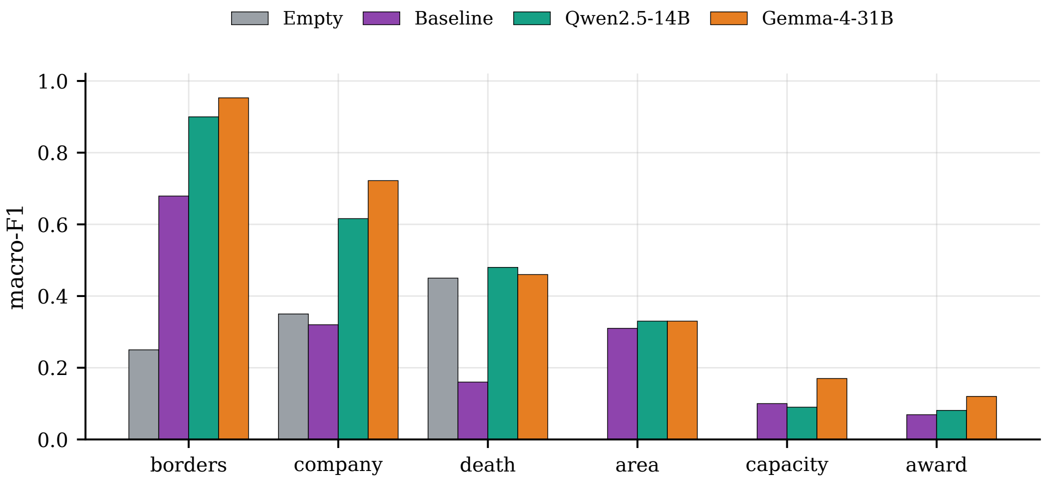

Figure 3: Per-relation validation macro-F_1: predict-nothing baseline, official baseline, Qwen2.5-14B, and Gemma-4-31B. Gemma-4-31B beats the official baseline on all six relations.

	Off	θ=0.5	θ=0.67
0.5B (avg of 3 null-heavy)	0.173	0.363	**0.372**
14B (avg of 3 null-heavy)	0.652	0.665	**0.666**

Table 4: Self-consistency abstention, relation-averaged over the three null-heavy relations only. "Off" decodes a single sample.

Threshold tracks the empty rate; numeric voting is neutral. If abstention drives the string relations, the natural next question is how to set its threshold. Figure 4 (left) sweeps θ: the most null-heavy relation, PERSONHASCITYOFDEATH (45% empty), peaks at a *lower* θ, while the others tolerate aggressive voting, arguing for per-relation thresholds. Figure 4 (right) varies the number of numeric samples: median aggregation is non-monotonic on HASAREA and actually slightly *hurts* HASCAPACITY (single sample 0.10 vs. median-5 0.09). Decode variance is not the bottleneck; the gap is knowledge-bound. The model's capacity guesses are the right order of magnitude but fall outside the 5% band (e.g. Spartan Stadium: predicted 37,000, gold 75,005).

AWARDWONBY is recall-capped. The one relation we beat the baseline on only narrowly exposes a different limit, and it constrains both models. Gold sets average ∼148 recipients (max 641), but in the 14B ablation the model enumerates only ∼11 names; its outputs are *not* truncated, it closes the JSON early, capping recall near $11/148 \approx 0.07$. The listed names are often plausible but wrong (P=0.36). Parametric recall and a reluctance to enumerate bound this relation, so a larger in-budget model or multi-step decomposition would help most.

Qualitative behaviour. Table 5 shows representative Gemma-4-31B outputs. The model is exact on well-known facts (PERSONHASCITYOFDEATH, HASCAPACITY) and near-complete on borders, but the two error modes are visible: a numeric near-miss on HASAREA, and on AWARDWONBY a list of globally famous names rather than the actual (obscurer) Stockholm honorary doctors, plausible but wrong, the recall/hallucination limit quantified above.

8 Conclusion

Every component of our system follows from one metric analysis: abstention pays off, numeric recall is scarce, and scope grounding curbs over- and under-prediction. With Gemma-4-31B it reaches 0.459 relation-averaged macro-F_1 (0.490 micro), beating the official baseline on all six relations, and is fully reproducible from a cached generation pass. A smaller Qwen2.5-14B, run locally in 4-bit, reaches 0.416. That the 30B wins with single-sample decoding, while abstention and self-consistency chiefly help the smaller models, suggests that model scale and decode-time calibration are complementary levers whose balance shifts as models get stronger.

Limitations

We report no test-set numbers yet, and the voting threshold θ is swept on the same validation

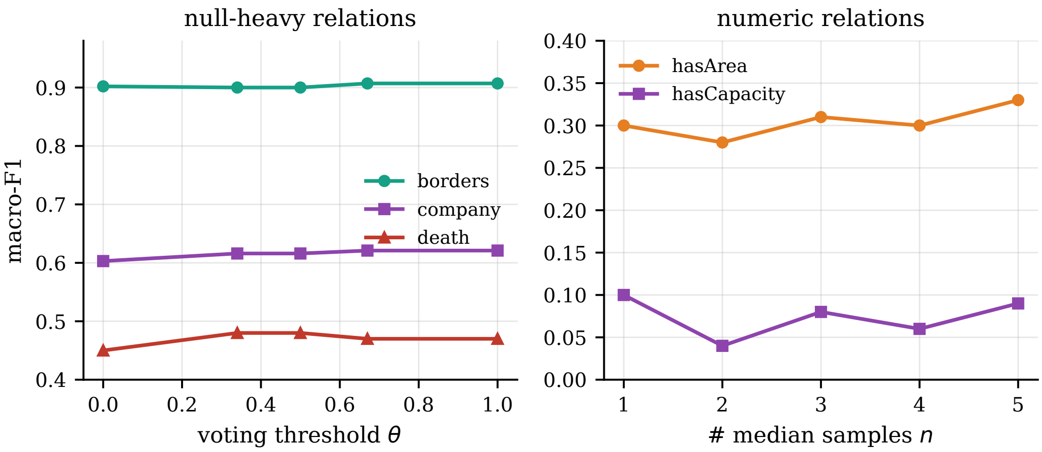

Figure 4: Left: per-relation F_1 vs. voting threshold θ (Qwen2.5-14B); the most null-heavy relation (DEATH, 45% empty) peaks at a lower θ than the others. Right: numeric F_1 vs. number of median samples n; aggregation is roughly neutral, indicating a knowledge-bound gap.

Relation	Subject	Gold (abbrev.)	Gemma-4-31B output	
borders	Ethiopia	Djibouti, Eritrea, Kenya, Somalia, S. Sudan, Sudan	Djibouti, Eritrea, Kenya, Somalia, S. Sudan	✓ 5/6
death	Vito Acconci	New York City	New York City	✓
company	Bharti Airtel	Bombay Stock Exchange	NSE of India; Bombay Stock Exchange	✓
capacity	Gileno de Carli	5000	5000	✓
area	Lošinj	74.36	123.3	×
award	Stockholm hon. dr	Torvalds, B. Andersson, Gates, . . . (40)	Mandela, Tutu, Annan, Yunus, . . .	×

Table 5: Qualitative examples: Gemma-4-31B vs. gold. Exact hits on PERSONCITYOFDEATH/HASCAPACITY, near-complete borders; the numeric near-miss (HASAREA) and plausible-but-wrong enumeration (AWARDWONBY) are the knowledge-bound errors of Section 7.

split it is evaluated on, so its per-relation optima are optimistic rather than held-out. Several margins are within plausible decoding noise. AWARDWONBY has only 10 subjects, and because we did not measure the run-to-run variance of the stochastic decoding (N=3, n=5), sub-0.03 deltas (e.g. the PERSONHASCITYOFDEATH +0.03) are indicative only. Being closed-book, the system inherits the base model's coverage and recency limits: it cannot recover facts the model never stored, which caps AWARDWONBY recall and rare-venue numeric accuracy. Because API rate limits precluded multi-sample decoding, our Gemma-4-31B scores are single-sample; pairing the 30B with the self-consistency and abstention voting that help the smaller models is left to future work and may further lift the null-heavy and numeric relations. Finally, the abstention advantage stems from this task's macro metric and its empty-as-valid-answer setting, so it may not transfer to single-answer or micro-only regimes.

Ethics and Generative-AI Declaration

Closed-book KB construction surfaces facts from model parameters that may be stale, biased, or hallucinated. Predictions about people (e.g. place of death) can be wrong, so they should not be treated as authoritative without verification. Our abstention mechanism partly mitigates this: when the model is uncertain it emits an empty answer rather than a confident fabrication. The only generative-AI component is the open-weight LM that produces the predictions, as the task requires; we used no proprietary assistant to generate results.

References

Dimitrios Alivanistos, Selene Báez Santamaría, Michael Cochez, Jan-Christoph Kalo, Emile van Krieken, and Thiviyan Thanapalasingam. 2022. Prompting as probing: Using language models for knowledge base construction. In *Semantic Web Challenge on KB Construction from Pre-trained Language Models (LM-KBC)*.

Badr AlKhamissi, Millicent Li, Asli Celikyilmaz, Mona Diab, and Marjan Ghazvininejad. 2022. A review on language models as knowledge bases. *arXiv preprint arXiv:2204.06031*.

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *ICLR*.

Tom B. Brown and 1 others. 2020. Language models are few-shot learners. *NeurIPS*.

Boxi Cao, Hongyu Lin, Xianpei Han, Le Sun, Lingyong Yan, Meng Liao, Tong Xue, and Jin Xu. 2021. Knowledgeable or educated guess? revisiting language models as knowledge bases. In *ACL*.

Xinyun Chen, Renat Aksitov, Uri Alon, Jie Ren, Kefan Xiao, Pengcheng Yin, Sushant Prakash, Charles Sutton, Xuezhi Wang, and Denny Zhou. 2023. Universal self-consistency for large language model generation. *arXiv preprint arXiv:2311.17311*.

Jeremy R. Cole, Michael J.Q. Zhang, Daniel Gillick, Julian Martin Eisenschlos, Bhuwan Dhingra, and Jacob Eisenstein. 2023. Selectively answering ambiguous questions. In *EMNLP*.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient finetuning of quantized LLMs. *NeurIPS*.

Xin Dong, Evgeniy Gabrilovich, Geremy Heitz, Wilko Horn, Ni Lao, Kevin Murphy, Thomas Strohmann, Shaohua Sun, and Wei Zhang. 2014. Knowledge vault: A web-scale approach to probabilistic knowledge fusion. In *KDD*.

Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. In *NeurIPS*.

Gemma Team. 2024. Gemma: Open models based on gemini research and technology. *arXiv preprint arXiv:2403.08295*.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *ICML*.

Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. 2020. REALM: Retrieval-augmented language model pre-training. In *ICML*.

Zhengbao Jiang, Frank F. Xu, Jun Araki, and Graham Neubig. 2020. How can we know what language models know? *Transactions of the ACL*.

Saurav Kadavath, Tom Conerly, Amanda Askell, and 1 others. 2022. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*.

Jan-Christoph Kalo, Tuan-Phong Nguyen, Simon Razniewski, and Bohui Zhang. 2024. Preface: Knowledge base construction from pre-trained language models (lm-kbc) 2024. In *LM-KBC Challenge at ISWC, CEUR Workshop Proceedings*.

Jan-Christoph Kalo, Simon Razniewski, Bohui Zhang, and Tuan-Phong Nguyen. 2025. LM-KBC 2025: 4th challenge on knowledge base construction from pre-trained language models. In *LM-KBC Challenge, CEUR Workshop Proceedings*.

Jan-Christoph Kalo, Sneha Singhania, Simon Razniewski, and Jeff Z. Pan. 2023. LM-KBC 2023: The 2nd challenge on knowledge base construction from pre-trained language models. In *Knowledge Base Construction from Pre-trained Language Models (LM-KBC), CEUR Workshop Proceedings*.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *ICLR*.

Patrick Lewis and 1 others. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS*.

Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. When not to trust language models: Investigating the effectiveness of parametric and non-parametric memories. In *ACL*.

Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. 2023. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *EMNLP*.

Jeff Z. Pan, Simon Razniewski, Jan-Christoph Kalo, and 1 others. 2023. Large language models and knowledge graphs: Opportunities and challenges. *Transactions on Graph Data and Knowledge (TGDK)*.

Fabio Petroni, Tim Rocktäschel, Sebastian Riedel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, and Alexander Miller. 2019. Language models as knowledge bases? In *EMNLP-IJCNLP*.

Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Pranav Rajpurkar, Robin Jia, and Percy Liang. 2018. Know what you don't know: Unanswerable questions for SQuAD. In *ACL*.

Adam Roberts, Colin Raffel, and Noam Shazeer. 2020. How much knowledge can you pack into the parameters of a language model? In *EMNLP*.

Taylor Shin, Yasaman Razeghi, Robert L. Logan IV, Eric Wallace, and Sameer Singh. 2020. AutoPrompt: Eliciting knowledge from language models with automatically generated prompts. In *EMNLP*.

Denny Vrandečić and Markus Krötzsch. 2014. Wikidata: A free collaborative knowledgebase. *Communications of the ACM*, 57(10).

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *ICLR*.

Bohui Zhang, Ioannis Reklos, Nitisha Jain, Albert Meroño-Peñuela, and Elena Simperl. 2023. Using large language models for knowledge engineering (LLMKE): A case study on wikidata. *arXiv preprint arXiv:2309.08491*.

Zexuan Zhong, Dan Friedman, and Danqi Chen. 2021. Factual probing is [MASK]: Learning vs. learning to recall. In *NAACL*.